

Who Owns Intelligence?

The Future I Envision

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**Personal AI will become
part of the computer.**

Useful intelligence, included in the next ordinary purchase.

Two enduring platform paths.
No winner takes all.

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Would you send your most private question to the cloud?

Would you send the question that matters most to you, including the details you least want anyone else to know, to a cloud AI?

Perhaps it concerns a difficult relationship, unpublished research, or a decision you cannot yet discuss with colleagues. Useful advice requires more than a few harmless facts. The assistant needs to know what happened, what you fear, and what you have not quite managed to explain even to yourself. As the conversation becomes more valuable, it becomes more revealing. Who then holds that understanding?

Anthropic's September 2026 threat intelligence report shows another side of the chat window. Investigators analyze account activity and content, connect patterns, and intervene against misuse. Its privacy guidance also permits designated personnel to review flagged material under controlled conditions. What appears to the user as a conversation can become a record that a service provider examines and acts upon.[1, 2]

Such records can enter real investigations. Reporting by the Palm Beach Post, based on case documents, described a former Goldman Sachs analyst who made violent threats against a former girlfriend in ChatGPT. OpenAI reported the threats to the FBI, and the chats became part of a police investigation that also involved stalking and direct threats. Preventing violence is not the same issue as supervising ordinary private thought. The fact relevant to every user is that an intimate interface does not remove the provider's access to the record.[3]

Payment does not confer complete control over supply either. On September 10, OpenAI paused new subscriptions and upgrades to ChatGPT Pro 20X while preserving existing subscriptions. Anthropic's biology safeguards for Fable route certain requests to a less capable model, with the triggering policy subsequently revised. Access, allowances and the capability actually delivered all remain mediated by a service's rules.[4–6]

The most powerful systems are also attracting work far beyond personal assistance. OpenAI's reported candidate proof for the Navier–Stokes problem involved approximately ten thousand concurrent agents and an 88-hour search, according to its account. The mathematical claim remains subject to independent review and recognition procedures. It illustrates expanding demand, not an established cause of subscription restrictions. Cheaper computation can invite people to buy much more of it for problems they could not previously attempt.[7, 8]

An individual needs a different kind of certainty. Yesterday's files should remain accessible tomorrow. Work should continue. Obtaining help should not require sending the most important background elsewhere first. Perplexity's Portable Computer has begun to turn this need into a product direction. Its local-first design for DGX Spark starts work on the owner's device and seeks outside assistance through authorized disclosures. Understanding private material can precede the decision to upload it.[9, 10]

I expect an even more ordinary route to matter most. Many people will not make a separate decision to buy an AI workstation. They will replace a computer for work, study and life. Improving models and software will allow a growing share of those purchases to include useful private intelligence.

Personal AI will become a standard capability of the computer, not only a service purchased separately.

A personal computer need not overtake a data center for this to happen. The intelligence required for a familiar task must enter the resources people are already buying. Hardware improves, models alter the work required, and software connects the two. Where those changes meet, personal AI can arrive through an ordinary replacement purchase or even a model update.

This changes the financial comparison. A work computer that someone would buy anyway does not become an entirely new AI investment when local inference is enabled. It changes the industrial question too. Delivering demanding models dependably and bringing sufficient intelligence into a broad range of general-purpose computers can support different platforms. I expect NVIDIA and AMD to become enduring poles in the open personal-computing market, rather than one winner across every use case. Apple retains a distinct route through its integrated devices and operating systems.

Understanding this future requires connecting three questions. What intelligence does a person need? What additional cost does ownership impose? Who can keep that capability useful over time? They are not three separate leaderboards. They concern the same computer, the same work and the same person's choices.

I. Useful intelligence is moving within personal reach

A new model is usually introduced by whom it has surpassed. Yet what the highest-ranked system can do is a different question from what we should use every time. Revising an email, finding a document or repairing familiar code first requires an adequate result, not evidence that the system behind it has the world's highest score.

A personal computer never needed to outperform a mainframe to deserve a place on a desk. It changed how computation was obtained. People could complete their own work on their own schedule without requesting remote resources for every step. Personal AI can develop in a similar direction. The cloud raises the ceiling while local systems absorb increasingly mature tasks. Their maximum capabilities do not have to converge.

I understand this as a moving Pareto frontier between capability and resources. A fixed budget can buy better results, and a fixed result can require fewer resources. For individuals, the decisive event is not movement at the frontier's highest point. It is a useful part of that frontier crossing everyday requirements.

30B is not a permanent capability level

The Qwen lineage offers an observable window. From Qwen1.5-32B, Qwen2.5-32B and QwQ through the Qwen3 branches and newer 27B and 35B models, capabilities have changed inside a personally deployable parameter range. This essay primarily follows general-purpose

models around 25B to 40B total parameters, with Gemma as a comparison. The range makes a history tractable. It does not prescribe the size of future personal intelligence.[11–14]

The Artificial Analysis series uses the same v4.3 index, retaining each record’s collection date and estimation flag. Five Qwen reasoning configurations not marked as estimates score 7, 10, 19, 22 and 34. Their total parameters have not increased by an order of magnitude, but their evaluated capabilities have changed substantially. An unsatisfactory answer from a past local model is not a verdict on every future model the same hardware might run.[15–20]

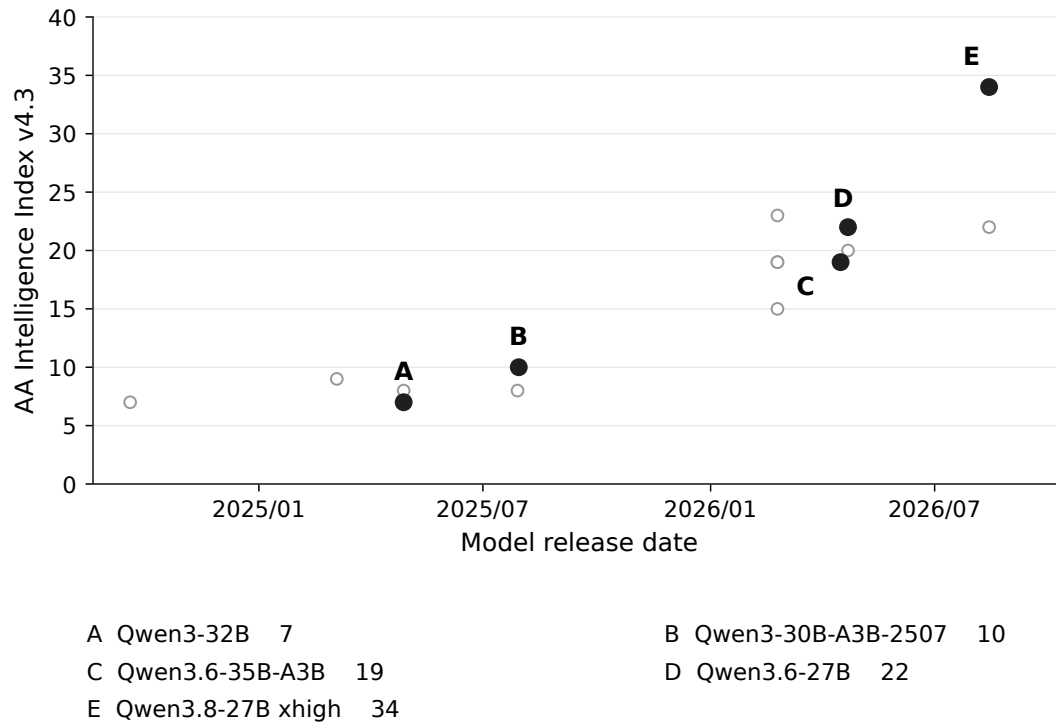


Figure 1. A similar deployment range does not imply a fixed capability level.

Dated AA v4.3 observations. Letters identify models beneath the chart; open markers indicate estimates. Architectures and reasoning effort differ. No score is assigned to an unreleased successor. Appendix A preserves the lineage.

Gemma points in the same direction. The comparable-size observations include 5 for Gemma 3 27B without reasoning, 15 for Gemma 4 31B with reasoning, and an estimated 17 for the 26B A4B branch. The change in mode is itself part of the change in performance. These observations show movement within a deployment range, not that intelligence has multiplied by the ratio between two index values.[21–23]

Sufficient also means worth waiting for

A higher score sometimes comes from more efficient computation and sometimes from more computation. AA’s release summary gives Qwen3.8-27B scores of 22, 26, 28 and 34 for non-reasoning, low, medium and xhigh settings. Choosing a model therefore also involves choosing how long it may work.[24]

A model can produce extensive reasoning before delivering a few hundred visible tokens. The answer looks short while the computation is long. Another configuration may answer quickly but require more correction. The quantity that should improve is the complete cost of an accepted result, not merely the length or speed of the first response.

A sufficiency threshold should consequently be anchored in stable tasks. A code change should pass the same tests. A document answer should recover the same essential evidence. Tool use should respect the same permissions. A new cloud model acquiring another skill should not automatically raise the passing line for every established task. Otherwise the claim that local intelligence need not be the best quietly disappears during evaluation.

The threshold for personal intelligence is not catching the leading model. It is completing familiar work well enough that remote assistance is no longer worth requesting every time.

Different tasks cross at different times. Transcription may remain nearby first, followed by file organization and eventually more complex coding and multi-step work. Retaining the cloud for difficult cases does not erase the value of work already done locally. But judging what can stay local requires examining what an agent actually asks a computer to do.

II. The whole job behind a short answer

Ask an agent to change a code repository. It reads files, proposes edits, runs tests and revises its approach after errors. Each new response may contain little more than a tool invocation. Whether the next step is correct nevertheless depends on earlier code, failed attempts and the latest result remaining available.

SemiAnalysis's AgentX collects voluntarily supplied Claude Code traces. The full corpus has median request lengths of 142,016 input tokens and 444 output tokens. Removing requests above 256K produces medians of 88,768 and 376. This is not a census of ordinary users, but it makes an important workload shape visible. Agents often read a great deal and produce only the next small step.[25, 26]

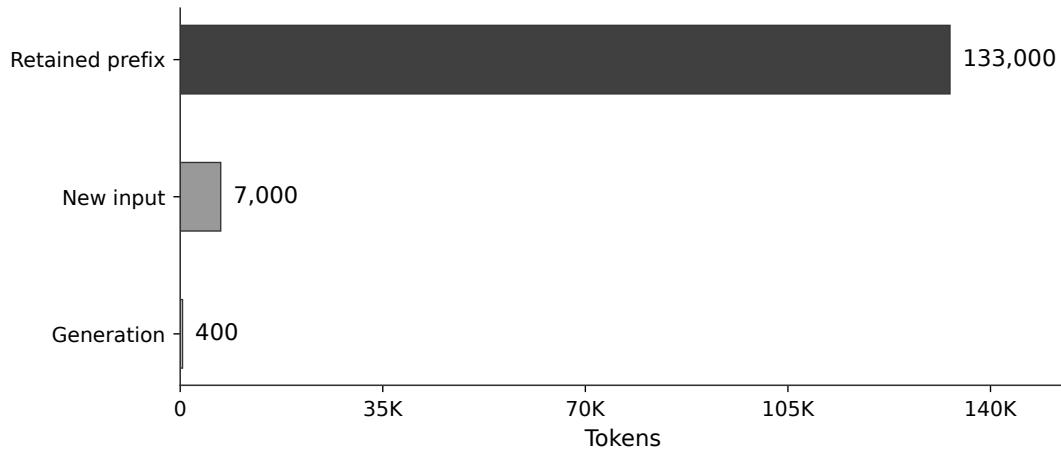


Figure 2. A few hundred new tokens can stand on a long history.

The analytical scenario has 140K total input, 95% device-resident reuse and 400 actual generated tokens. A 133K retained prefix, 7K new suffix and output are shown separately. This is not an observed joint median.

The analysis retains 140K input, 95% resident reuse and 400 generated tokens per stream as a demanding comparison point. Each call processes a 7K suffix after a retained 133K history, then generates the next step. Actual reasoning belongs inside the generation allowance. If a local model needs additional thought to succeed, those tokens must be restored to the account. AgentX's earlier 99.2% figure is a theoretical prefix-reuse ceiling; 95% is a separate residency assumption used here.[27]

A cache hit avoids repetition by retaining state

Input processing is called prefill, and generation is called decode. When a prefix recurs, its previously computed state can be reused. Yet a new 7K suffix after a long history is not an isolated 7K document. It must still be interpreted against the existing background.[28, 29]

KV cache is part of that background. It retains previously computed keys and values, saving repeated work while occupying memory. Moving state to host memory or an SSD can preserve more sessions but may delay their return to active use. A long-term document index, a recoverable conversation state and KV cache participating in the next computation are different resources.

Reuse across turns is also different from sharing across requests. Two agents branching from identical code and instructions can share an opening prefix. Agents working on unrelated repositories may retain separate histories. Hybrid architectures additionally require the recurrent state associated with a reusable position or a way to reconstruct it. Hits, sharing and residency must be measured separately to establish what was saved.[30, 31]

One person can still create concurrency

Agents sometimes wait for a terminal or network and sometimes return together with new results. Eight open sessions, eight retained histories and eight executing model requests impose different pressures. Batching can amortize weight reads, but it does not eliminate each independent history. An MoE batch may also touch a larger union of experts than a single token does.[32]

This separates a personal service from a demonstration. A foreground request needs a timely response while background work continues. A long prefill should not monopolize the device and interrupt every ongoing answer. Nor should responsiveness mean that background jobs never run. Scheduling, state management and computation jointly determine the experience.

Waiting compounds along a task's dependency chain. Forty serial model calls of twenty seconds each would account for more than thirteen minutes before tool time. The first visible token and the completion of useful work are therefore different events. Fast generation alone does not establish a fast agent.

The strict comparison asks for generation to begin within ten seconds, a 400-token call to finish within twenty, and each stream to sustain at least 30 token/s. This defines one demanding service, not the admission standard for all personal AI. A second valid test allows improved retrieval, shorter unnecessary prompts and tools loaded on demand, while preserving the same final task requirements. What must survive is useful work, not every token inherited from a cloud-oriented workflow.[10, 25]

My own partial estimate is about 500 calls daily through the interfaces I count, excluding some other routes such as Codex. It is a personal reference, not a market average. It supplies scale for the economic examples without establishing that all those calls have 140K input or that a local candidate can replace them all.

The system requirements are now concrete. Preserve context, absorb new information, produce the next step and keep several chains moving. Better hardware and better models must demonstrate their value inside this complete job.

III. The model changes the machine we need

The same job can be completed in different ways. Personal AI cost depends not only on operations per second, but on what a model asks hardware to store and read, and how much repetition software removes. Separating those demands explains why an unremarkable-looking computer can acquire a substantial new use.

Capacity and bandwidth solve different problems

A model must fit before it can run quickly. Capacity determines whether weights, history and workspaces can remain resident. Bandwidth determines how promptly their data reaches

computation. Consumer GPU families show different trajectories. RTX 3080, 4080 and 5080 provide 10GB, 16GB and 16GB of memory with 760, 716.8 and 960 GB/s of bandwidth. RTX 3090, 4090 and 5090 provide 24GB, 24GB and 32GB with 936, 1008 and 1792 GB/s.[33–38]

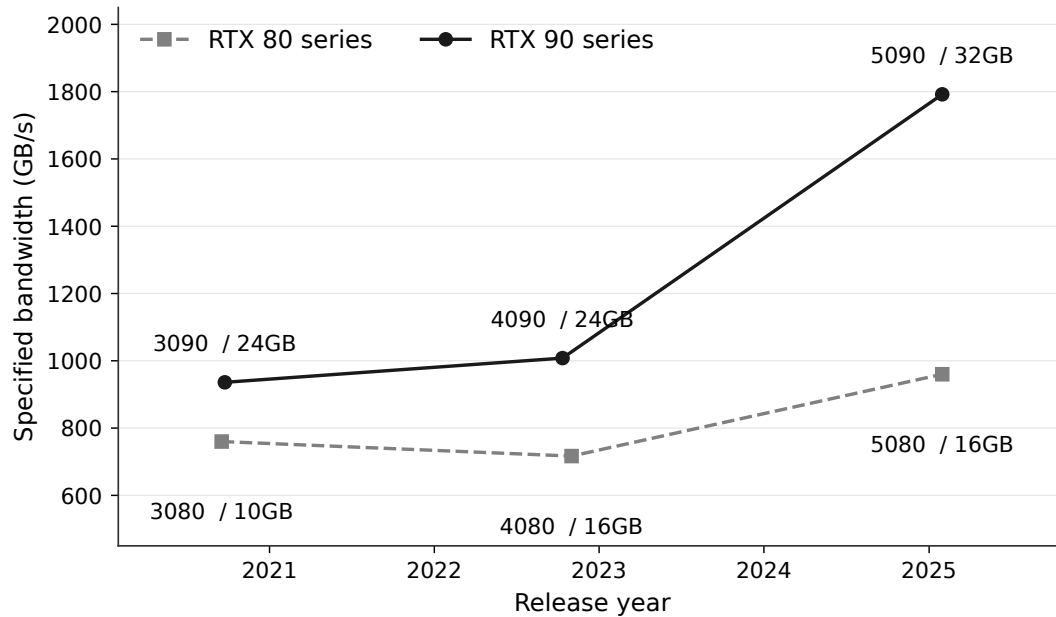


Figure 3. Capacity and bandwidth have not followed the same trajectory.

Desktop base models, excluding Ti, SUPER and laptops. Labels give model and memory capacity. Lines connect released specifications, not continuous measurements. Appendix A retains the historical data and fits.

Descriptive fits to the three generations imply annual bandwidth growth of approximately 5.6% for the 80 series and 16.3% for the 90 series. Yet fitting only the first two generations underpredicts the third by about 30% and 39%. Apple Max bandwidth also remained flat across several generations before moving from 400 to 546 and 614 GB/s. Interfaces, packaging and product positioning produce jumps. A smooth fit is not a specification promised to the next buyer.[39–43]

CPU improvements help a different part of the workload. Data preparation, tools, serial interaction and parallel compilation all use them. The selected same-version retrospective series gives descriptive annual single- and multi-core growth of approximately 11.5% and 17.3% for base Apple processors, and 12.7% and 13.9% for an eight-core AMD comparison. These characterize particular histories. They are not conversion factors from CPU scores into decode speed.[44, 45]

Consider the 27B reference. One public NVFP4 export contains about 20.42 GiB of tensor data. A 140K input plus 400 generated tokens requires approximately 4.28 GiB of FP8 KV payload per stream. Adding recurrent states and runtime reserves brings even one long session close to a 32GB card’s limit. Greater bandwidth cannot provide the missing space for another independent history.[46, 47]

First reduce the history that must be retained

Architecture is already changing this account. Qwen3-32B uses full attention in all 64 layers, adding 256 KiB of BF16 KV per token. Later hybrid 27B configurations retain full attention in 16 layers and update recurrent state elsewhere, reducing the growing term to 64 KiB per token. FP8 storage subsequently reduces it to 32 KiB. The architectural saving precedes quantization.[47–50]

Recurrent state does not grow in the same way with a single history, but prefix recovery and speculative execution can require additional copies. Hybrid attention reorganizes history rather than eliminating all state. This also separates the two costs of long reasoning. More thought takes more generation steps and can expand the history accessed afterward. Hybrid architecture reduces the latter without automatically shortening the former.

The released Qwen3.6-35B-A3B further separates total capacity from active work. Its language model has about 35B parameters with approximately 3B active per token. Ten of its 40 layers use full attention and thirty update recurrent states. The BF16 KV growth term is 20 KiB per token. Three independent histories of 140400 tokens therefore occupy about 8.03 GiB of raw KV.[14, 51]

Combining a roughly 22.4GB four-bit weight-file proxy with explicit state and workspace allowances gives about 35.3 GiB on the accelerator and 47.3 GiB for the whole host at three streams. A 64GB computer becomes a serious candidate. Eight streams approach 63 GiB in total, leaving little room and potentially exceeding the graphics allocation first. Actual accelerator availability matters as well as the memory printed on the box.[52, 53]

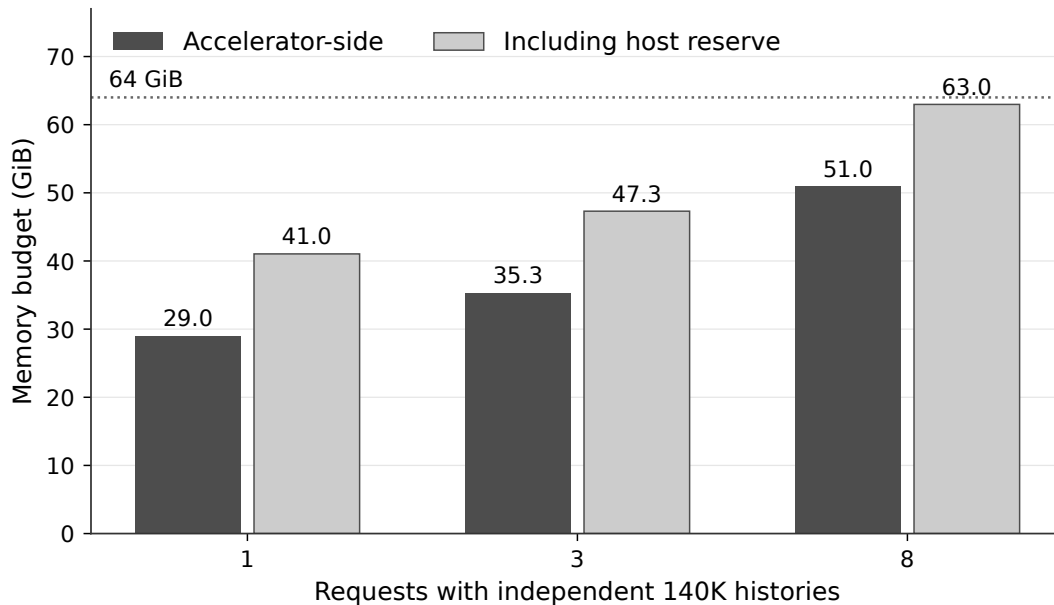


Figure 4. A compact active model can retain several long histories.

Constructed Qwen3.6-35B-A3B budget with BF16 KV and five FP32 state copies. Grouped bars separate device-side and whole-host requirements, including explicit runtime reserves. The 64 GiB line is a capacity reference, not guaranteed accelerator allocation. These are not loading measurements. Appendix B provides formulas and assumptions.

A suitable model changes what existing hardware can do

Framework’s Strix Halo results make the distinction visible in speed. Its approximately four-bit Qwen3.8-27B configuration processes about 391 input token/s and generates 11.6 token/s in a single stream. Qwen3.6-35B-A3B reaches about 1345 and 52.1. These are short-input results on the same device family, not equal-quality task comparisons. They nevertheless show that model selection can matter as much as whether the chip has changed.[54, 55]

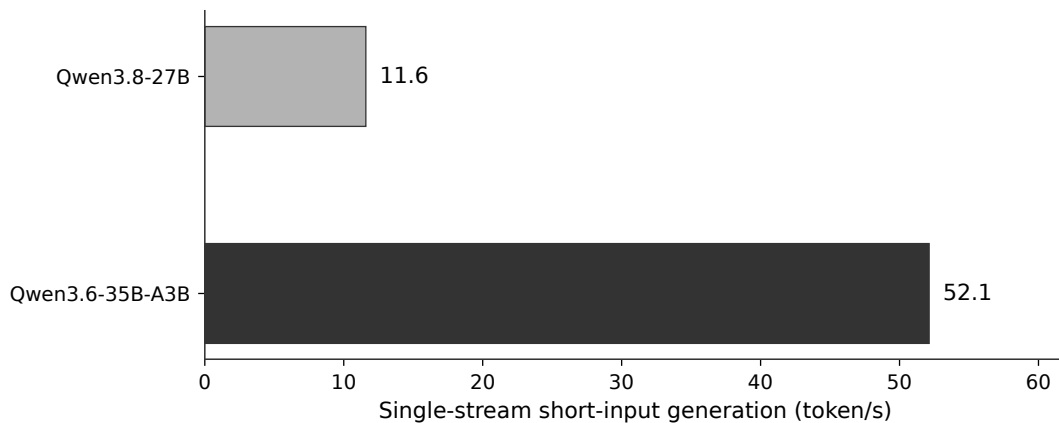


Figure 5. A different model can change the experience on existing hardware.

Framework single-stream short-input decode on its disclosed Linux, llama.cpp and ROCm configuration. This is neither generation at 140K depth nor a benchmark of the GMKtec 64GB machine. Input-processing results and provenance appear in Appendix B.

Prices are also entering an ordinary computer-buying range. GMKtec lists a Ryzen AI Max+ 395 system with 64GB memory and a 1TB SSD at USD 2199.99. Framework supplies a 64GB configuration from the same processor class. These are desktops, not promises about the battery life or sustained performance of a similarly priced laptop. They show that large unified memory need not be confined to a costly specialist station.[56, 57]

Spark embodies another sensible choice. It supplies 128GB coherent memory and 273 GB/s bandwidth for larger sparse models. Flash Next has approximately 125B main-model parameters, about 6B active per token, and additional lookup parameters suitable for different storage arrangements. Its design separates model capacity from per-step access, while selective attention changes the computation over long histories.[58–60]

One deployment using NVIDIA’s low-precision export loads about 76.48 GiB on the device while keeping approximately 47.68 GiB of lookup data on local NVMe. Other single-Spark records report around 91.9 token/s aggregate for three short-input streams, alongside separate long-input and sustained-use observations. Spark is already competitive with a suitable model, not merely capacious. The separate measurements do not establish a complete three-stream 140K acceptance test.[61, 62]

The purchase decision returns to the job. General-purpose PCs can retain work that compact models perform reliably. Users needing stronger models, more persistent state or continuous

background service may value a private station. These are simultaneous product segments, not an upgrade ladder that everyone must climb.

Connect the improvements into one system

Weight quantization reduces resident data and some repeated weight traffic. Activation quantization can put input processing on a more efficient arithmetic path. KV quantization reduces history storage and reads. Fused kernels and suitable layouts determine whether these savings survive execution rather than being spent again on conversion. Joint design allocates precision and resources within one workload. It cannot be represented by multiplying the best speedup in several unrelated papers.[63–66]

Speculative decoding addresses another expense. A cheaper drafter proposes tokens for the target to verify together, reducing costly serial rounds when acceptance is favorable. The benefit depends on acceptance, verification and concurrency. Once batching already amortizes weight reads, more drafting may become less attractive. Correct sampling preserves the distribution of the actual target, not the quality of a different, unquantized model.[67–69]

Compressing history must also preserve useful work. In one Qwen3-30B-A3B experiment, aggregate MRCR AUC over 8K to 256K falls from 45.8% with BF16 KV to 42.3% with a protected K4V4 configuration. Other 4K to 32K task records show much smaller differences. Reliability depends on model, task, length and precision allocation, not only bit width. Four-bit KV is valuable when it enables more dependable work within the same memory, not merely a smaller state file.[70, 71]

Adaptation after release is beginning to inform design before release. NVIDIA has publicly examined model shape, attention, hardware utilization and interaction targets together. Qwen running through open software on AMD shows that productive coordination can also cross company boundaries. Perplexity's changes to prompts, tool loading and post-training bring the application into that process as well.[10, 72–74]

Co-design changes not only how quickly the machine runs, but how much machine the same job requires.

One architecture can reduce the need for expensive bandwidth. Another state layout can lower capacity requirements. Reliable execution can let an ordinary device perform work previously reserved for a specialist system. The improvements need not arrive together or belong to one company. When they meet, the next leap in personal AI can be a model update.

IV. Personal AI will arrive with the next computer

Imagine an ordinary replacement purchase. The old computer has served for several years. Its owner needs better battery life, more memory or a screen better suited to work. The

purchase would happen whether or not local AI were used. The new machine, however, can do something more. It organizes files, transcribes conversation and perhaps handles familiar coding tasks.

This person has not decided to become a compute operator. No server was purchased solely to answer a few questions. A normal computer was bought and an included capability became useful. I expect this route to matter more for mass adoption than a separate personal workstation in every household.

A familiar price tier can contain more capability

Technical progress does not always appear as a large annual reduction in a product's price. It can sustain a recognizable price tier while changing what is included. Apple's US 14-inch professional tier started at USD 1999 with M1 Pro in 2021, M2 Pro in 2023 and M4 Pro in 2024. The example shows that product tiers can be more stable than configurations. It does not imply that all computers cost this amount or that a stable nominal price is a stable real cost.[75–77]

A plausible evolution follows. More memory, suitable processors and local model support become standard within some general-purpose tiers. Buyers still choose a development environment, display, weight, battery life and reliability. Useful intelligence becomes another part of that bundle.

When comparable products all contain the capability, declining an AI feature need not produce a meaningful hardware refund. A user can leave it disabled without having access to an otherwise identical machine that removes only AI and returns a large part of the price. This is a plausible standard-feature scenario, not a claim that cheaper or used computers disappear. A lower tier remains a valid alternative whenever it meets the buyer's underlying requirements.

Apple supplies an existing example at a smaller scale. Its Foundation Models framework exposes an on-device model through the operating system, with eligible local inference available offline and without a per-token charge. This does not certify a 140K coding agent. It demonstrates that intelligence can enter as a capability of an existing device and system rather than a remote call purchased anew each time.[78]

First one task, then several things at once

The route does not require every workload to mature simultaneously. A single everyday assistant needs dependable results and tolerable waiting. Two or three concurrent tasks add retained state and simultaneous input processing. Eight continuously active workloads test capacity, throughput, cooling and scheduling together. Such work may be better placed on a station than on the laptop serving the foreground user.

Compact MoE capacity and short-input speed make the first two levels worth testing on existing machines. Important long-history gaps remain. One public 128GB Strix Halo record

processes a cold 131072-token input at about 435 token/s. The short-input 1345 token/s result cannot substitute for it. That record's input and generation commands are also separate tests; its generation number is not automatically generation at the end of the long input.[79, 80]

The missing performance can be expressed as a service requirement. If each stream generates at 50 token/s at the required history depth, three streams receiving independent 7K suffixes need about 2339 aggregate suffix token/s and 150 aggregate generation token/s to meet the stated timing budget. Eight streams require about 6236 and 400 respectively. Fitting state, generating quickly and absorbing concurrent new input must hold together.

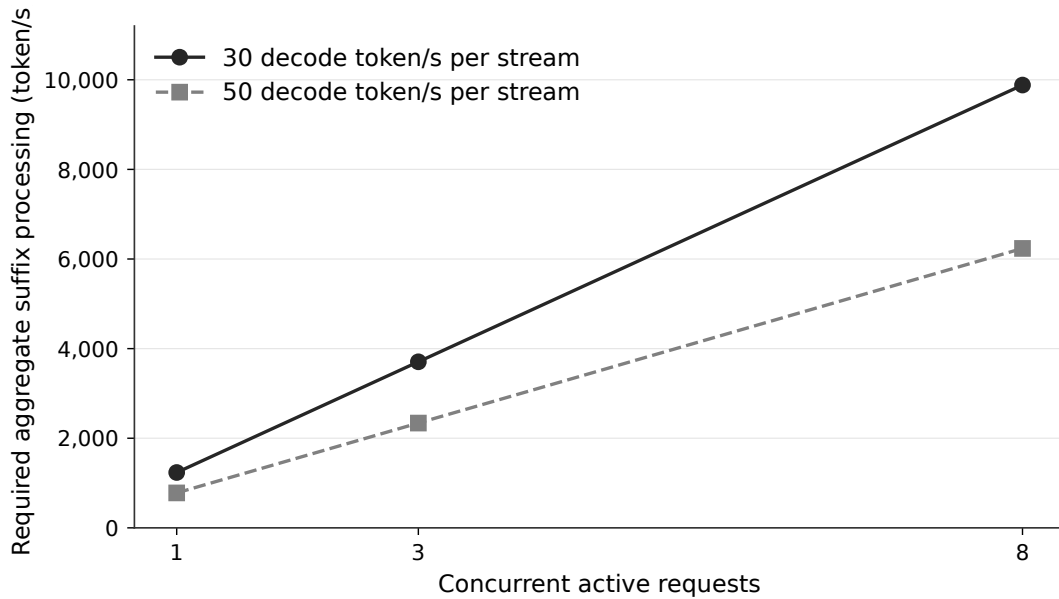


Figure 6. Concurrency is a service requirement, not a count of open windows.

Each stream has 140K input, 95% resident reuse and 400 generated tokens. With one second of extra overhead, the required aggregate suffix rate is derived from ten-second first-token and twenty-second completion targets. Lines describe requirements, not measured devices.

The arrival date belongs to the system that passes the task

A model release can let installed hardware cross these requirements. Runtime updates can then improve latency, while normal purchases spread the capability. Personal AI will not arrive on one universal hardware launch date. The useful object to track is a stable set of tasks and the purchasable combinations that can complete them.

Infrastructure tests can preserve a fixed 140K request trace. Product tests can permit better retrieval and workflow organization to complete the same task with less context. The first fixes the input; the second fixes the result. Restricting attention to the first can miss the application design that makes an ordinary computer sufficient. Using only the second without counting retries and outside help can hide unfinished work.

I do not treat the depreciation schedule of a USD 4699 machine as the beginning of the entire market. Near-USD-2000 PCs may complete some tasks first, with larger systems carrying

heavier work. The central prediction concerns this diffusion mechanism, not an assigned release date for an unpublished model.

Replacement turns technical capability into an installed base

A feature becoming possible and many people owning and using it are different events. Consider a stable population in which each computer has a one-in-six probability of replacement in each year, independent of age. If every new purchase now passes a fixed task, approximately 66.5% of devices become capable after six years. A four-year mean interval spreads capability faster and an eight-year mean interval more slowly. Updates reaching existing devices can shorten the delay, while not every capable owner will activate the feature.

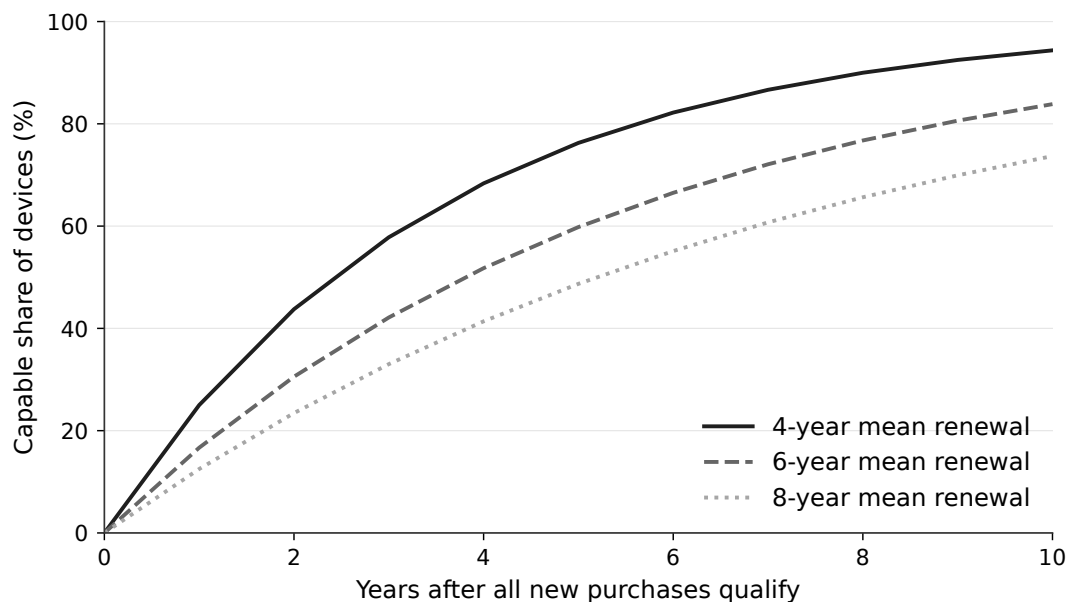


Figure 7. A technical threshold can spread through ordinary replacement.

Stable-population scenarios beginning when all new purchases qualify for a task. Older devices do not gain capability by software in these curves and renewal is uniform. The horizontal axis measures years after that event, not forecast calendar years. Capability is not active use.

Longer useful life improves ownership economics but can slow the spread of new hardware capabilities. Software that improves older devices allows the two objectives to coexist. Research should therefore track the share of genuinely capable systems in ordinary purchase tiers and the existing machines upgraded into eligibility, not only shipments of dedicated AI stations.

Mass adoption need not begin with a new capital-investment decision made specifically for AI. Useful local intelligence can be a new capability of a purchase that would happen anyway.

That is the economic starting point. The alternatives are not simply a computer and a cloud bill. They are the buying and working arrangements a person would otherwise choose, and what local intelligence adds to or subtracts from them.

V. Only the additional investment has to pay for itself

One person is replacing a computer. Another is adding a Spark beside an existing one. Both may gain local AI, but their economic decisions differ. The first asks what a more suitable configuration adds to an existing purchase. The second asks whether an entire additional device is worth buying. Combining their accounts obscures both the opportunity for mass adoption and the hurdle facing dedicated equipment.

Begin with the purchase that would happen without local AI

Cloud use also requires an access device. Someone who develops software, handles documents and attends video meetings cannot remove a work computer from their life merely by renting inference. The money genuinely available for cloud services is the difference between two acceptable purchasing arrangements.

Suppose a buyer would otherwise choose a USD 1800 computer and instead buys a USD 2000 configuration for local AI. With the same timing, lifetime and terminal value, the additional purchase attributable to AI is USD 200. If both alternatives use the same USD 2000 machine, enabling local inference adds no purchase expenditure. An otherwise unplanned USD 4699 station, in contrast, belongs in full on the investment side.

This is not an arbitrary allocation of a computer's price among uses. It compares actual alternatives. Display, battery life, weight, development environment, warranty and non-AI performance must remain acceptable. A cheap terminal cannot simply replace a work computer by assumption. Nor does a buyer's historical USD 2000 budget become immutable. A cheaper configuration that meets the real requirements is a legitimate alternative.

If local intelligence becomes standard within a tier, the identifiable configuration premium can be small or zero. The manufacturing cost still exists and may be embedded in everyone's price. But a person choosing between actual arrangements should not be charged for a second purchase that does not occur. A bundled general capability and a dedicated appliance have different counterfactual costs.

An asset can keep providing work

Even a dedicated device is not a service fee consumed on payment. An expired subscription supplied past access. A computer used for a year can continue working, change roles or be sold. Capital services, market value and the date of retirement are different concepts.[81]

RTX 4090 launched on October 12, 2022, almost four years before this essay's reference date. Subsequent models and software can still expand supported workloads on installed RTX hardware. Spark has also received model and execution updates after release. A machine may become more useful while losing resale value, or remain physically functional after becoming unsuitable for a demanding task. The next product launch alone does not determine its economic life.[82–84]

With six years of use, zero terminal value and an 8% annual real capital rate, monthly capital recovery is USD 34.79 for a USD 2000 purchase and USD 81.75 for USD 4699. A USD 200 configuration increment costs only USD 3.48 under the same assumptions. These amounts already include capital opportunity cost and must not be charged again as separate depreciation or interest. Appendix C retains three-, four- and eight-year cases and residual sensitivities. Six years is a scenario, not a promised hardware lifetime.

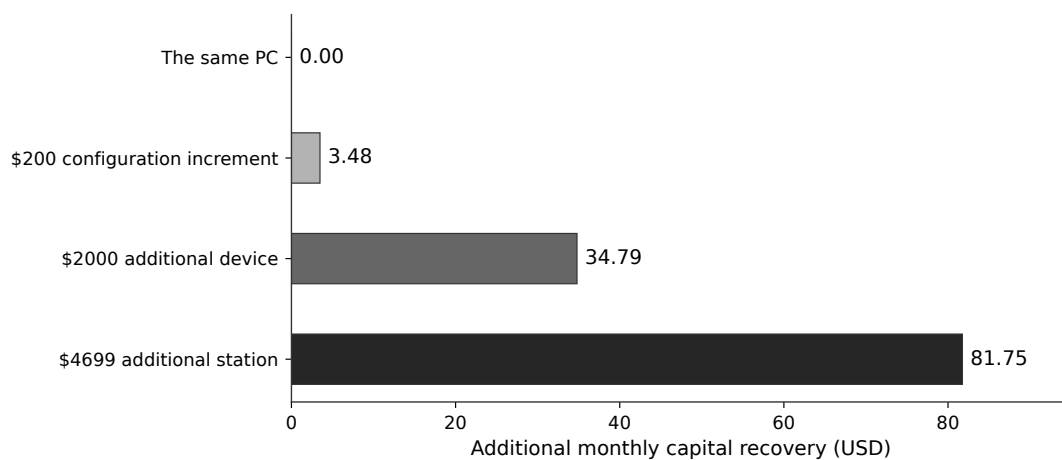


Figure 8. The additional capital can differ greatly for the same decision to use AI.

Six years, zero terminal value and an 8% annual real capital rate. The first two cases are normal-purchase counterfactual differences; the last two are additional devices. Capital only, before operating and opportunity costs.

Local execution is not free, but its added cost can be small

The normal-replacement account includes additional electricity, maintenance, applications and resource contention. Keeping a computer awake when it would otherwise sleep adds power. Slower compilation, gaming or foreground response has an opportunity cost. Buying earlier specifically for AI changes the timing of expenditure and may forgo remaining value in the old device.

The following examples compare five purchases. They assume six-year use, zero residual, an 8% real capital rate, four hours of initial setup and fifteen extra minutes of monthly maintenance valued at USD 50 per hour, without an additional application subscription. An annual repair allowance equals 2% of added hardware investment. For a shared PC already powered during the relevant use, inference adds an assumed 100W and twenty allocated device-seconds per call. Dedicated systems instead include 40W idle, 200W active and 730 available hours monthly. Electricity costs USD 0.20 per kWh.

Actual purchase difference	Added capital	Added monthly cost
The same normally purchased PC	0	17.65
Configuration increment at replacement	200	21.46
Configuration increment at replacement	500	27.18
Additional qualifying system	2000	62.61
Additional dedicated station	4699	114.07

Table 1. Identify what was added before asking what it costs.

USD at 15000 eligible base calls completed locally each month. Power, labor, repair and life are scenarios, not measurements. The shared PC does not displace other work in this table; actual contention must be added. Equal service across the devices is not established by the accounting.

The shared cases neither erase operating costs because a PC exists nor charge its full purchase again. The dedicated cases carry their complete additional capital and availability cost. The resulting conclusions can coexist. A separate station may not pay for itself while local execution on the next normally purchased PC is worth enabling.

The economical model belongs on the cloud side too

For Spark's larger-model route, the related rental alternative is the official Flash service based on Flash Next. Qwen documents the relationship, but the production endpoint is not guaranteed to be identical to a local NVFP4 file. Compact local systems must similarly face the cheapest eligible task-level alternative, not an expensive API selected to make ownership appear cheaper.[85]

The September 15 official Singapore tariffs per million tokens are USD 0.150 for ordinary input, 0.016 for cache reads and 0.470 for output. Virginia's corresponding rates are 0.113, 0.014 and 0.382. At 140K input, 95% billable cache hits and 400 output tokens, 15000 calls cost USD 50.49 and USD 42.09 in model tokens. Multiplying my partial estimate of 500 daily calls by thirty days supplies a volume scenario, not my actual invoice.[86, 87]

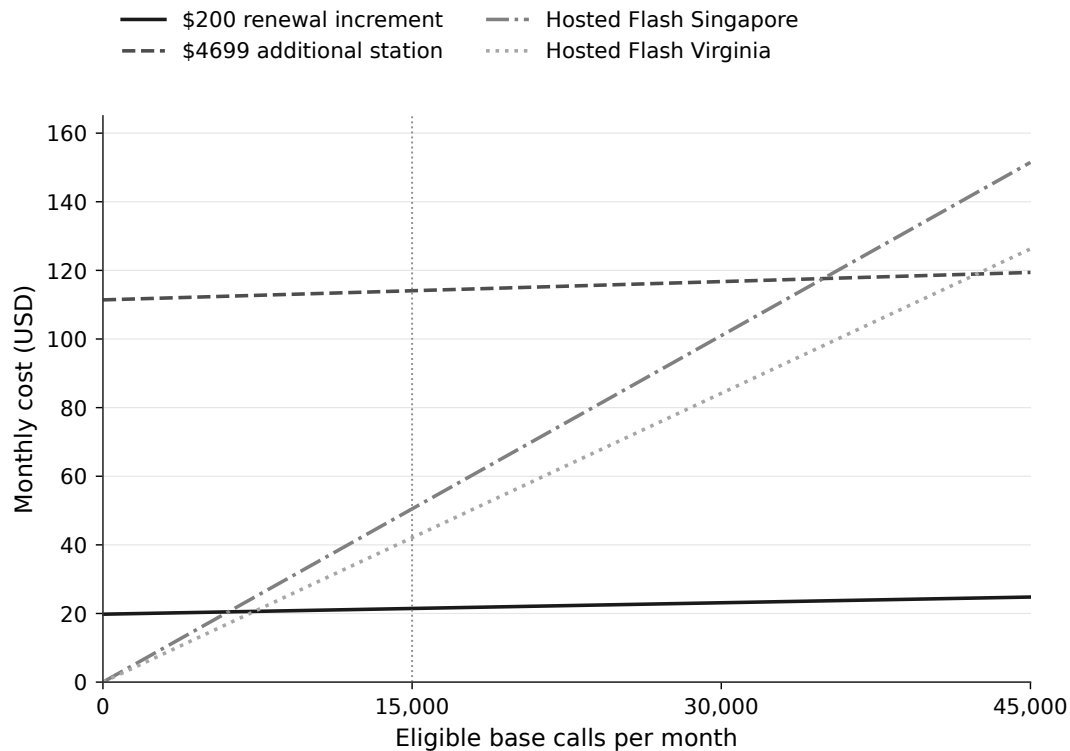


Figure 9. Normal replacement and an additional station have different break-even curves.

Six-year low-friction ownership and dated Flash token prices. The shared case charges a USD 200 configuration increment; the dedicated case charges the full device. Calls must qualify at equivalent task quality. Common client costs cancel and noncommon costs are additional. The vertical line is a personal volume scenario.

Under these conditions, a dedicated Spark-scale purchase at about USD 114 monthly remains more expensive than the USD 42 to USD 50 cloud token bill. A USD 200 configuration increment at about USD 21 monthly lies on the other side. The difference does not mean a household has surpassed datacenter production efficiency. It means useful work is being added to an asset purchased for other purposes.

The cloud's scale advantage remains real. It can pool demand, improve utilization and spread purchasing and operations across users, while customers can reduce rental when they need less. Quantization and efficient models benefit it too. The comparison should face the actual service the user can buy. Neither ignoring scale nor assuming that a provider's lowest production cost is passed to every retail customer answers that question.[88, 89]

The work that can stay local matters more than the total call count

Fifteen thousand calls do not imply fifteen thousand local replacements. Cash savings require the local model to pass the task and reduce an actual metered charge, overage payment or subscription tier. An unchanged Codex or other subscription has not been saved.

In the USD 200 configuration-increment case, retaining half the calls at equal quality costs about USD 20.63 monthly in additional local expenses. Avoided Singapore and Virginia model bills are USD 25.25 and USD 21.04. The remaining calls still use the cloud and cancel

between the alternatives. The monetary margin is narrow. Another half-hour of maintenance or necessary correction can reverse it.

At static tariffs, approximately 6081 eligible local calls a month are required against Singapore, around 41% of 15000 calls. Virginia requires about 7345, or 49%. These are derived thresholds, not predictions of a small model's replacement ability. A better model improves economics by increasing the share of work reliably retained, not merely by shortening a particular response.

Caching and outside advice also require symmetric treatment. Common adviser calls cancel when both routes need the same assistance. Differential failures, remote cold starts and longer reasoning remain costs. A 95% local residency rate does not establish 95% cloud billing hits. Lower cloud cache reuse would increase the reference bill, but that difference must come from traces rather than an assumption that remote caches always fail.

From cheaper to worth owning

A six-year asset competes with six years of cloud progress. Payment in the past does not make cheaper alternatives irrelevant, and a cheaper future computer does not refund today's invoice. Discounted comparisons should hold the purchased arrangement fixed while following useful work, operating costs and eligible cloud prices. Model updates can extend usefulness and enable new private work. Calls that would never otherwise have been purchased are benefits, not automatically avoided cash expenditure.

Privacy and control can independently justify a purchase. A person can accept an explicit premium to retain original material or exclude services that fail a confidentiality requirement. No universal dollar value for privacy is required. Local permissions, backups and security still deserve the same scrutiny. Physical possession does not ensure reliability.

A dedicated station must justify an entire additional machine. Personal AI on a general-purpose PC must justify the additional cost of using it. The same inexpensive cloud can produce different answers to those two decisions.

Mass adoption need not persuade everyone to operate a cheaper miniature datacenter. It can make devices already being purchased perform useful private work at a modest added burden. The platform problem then moves from supplying more computation to requiring less attention from the owner.

VI. A default capability must become a dependable way of working

A feature included in a computer has not necessarily entered everyday life. People need a system that still works tomorrow. Files remain available, updates preserve permissions, and background agents do not disrupt a meeting. Maintenance time is easy to omit from an incremental-cost model and precisely the burden a product must actually remove.

Perplexity's Portable Computer demonstration has a local agent read private documents, cite pages and save results. That workflow is closer to value than a generation-speed number. Original material becomes the beginning of local work, with external information or stronger models added when needed. Computation has been organized into something a person can finish.[90]

Portable devices sense while personal systems retain context

Personal AI need not occupy one machine. Phones, earbuds and glasses can handle wake words, voice activity detection and transcription. Specialized speech models are generally much smaller than 30B and can be selected for latency, language, noise and battery requirements. Raw sound can first become text where it originates, before deciding what needs further processing.[91]

More demanding work moves to a personal PC or station holding authorized files, memory and task state. Changing screens need not move the whole private background into another provider's service. Persistent intensive jobs can occupy a separate station while routine interaction stays on an existing device. Whether another machine is warranted depends on time, capacity and the cost of interfering with other work.

The cloud remains important, but who decides when it is used?

A local system can seek advice from a remote model. Perplexity's published advisor design selects context, flags sensitive material and previews what will be sent. The remote assistant receives authorized material and returns guidance while local tools and file execution remain under local control.[10]

Removing names is not complete anonymization. Dates, locations, unusual experiences and precise figures can reidentify a person. A better operating principle is minimal disclosure, with a visible outbound boundary and the option not to upload particularly sensitive work. Disliking an answer can request another attempt or a suggested escalation. It should not silently authorize additional disclosure.[92]

Search queries and external connectors also transmit information. Local execution is not the same as never using a network. Subscriptions may remain too. Portable Computer's launch arrangement serves paying subscribers; the absence of local per-token charges does not make every application capability permanently independent of a subscription. Device ownership, model rights and application services are separate relationships.[9, 93]

A dependable system should allow original material to be exported, models replaced, versions pinned, permissions withdrawn and work recovered. Deleting, paying and sending need explicit action boundaries. Apple's Private Cloud Compute offers another comparison through specialized execution and verification intended to limit provider access. The relevant question is not whether local or remote computation deserves an absolute safety label. It is what the user can verify, decide and retain.[94]

These requirements also explain why a platform can create value. An ecosystem that persistently reduces installation, maintenance and migration effort gives people a reason to stay. Retention comes from continued usefulness rather than custody of their files. This supplies the standard for judging the industry. Whoever organizes a durable relationship can earn value, but different users may need different relationships rather than a single organizer.

VII. Two platform paths rather than one winner

When personal AI arrives through normal computer purchases, the industrial question is larger than who sells the most dedicated stations. It is who can turn changing model capabilities into dependable features across purchase tiers. Some users pay for stronger execution and clear support. Others first need enough memory and adequate capability within a familiar budget. Those needs need not converge on one product.

My base case is persistent coexistence rather than winner-take-all. NVIDIA and AMD become enduring poles in open PCs and independent personal-computing nodes, while Apple retains a strong integrated route inside its own device ecosystem.

In the language of mobile platforms, NVIDIA is more likely to resemble the iOS-style proposition of tightly coordinated models, execution and support. AMD is more likely to support an Android-style breadth of distribution, with manufacturers combining hardware and open software across more price points. The analogy concerns integration and distribution, not ownership of two phone operating systems or the claim that every AMD component is an open standard.

Two poles do not require equal shares or exclude other suppliers. Gaming GPUs offer a history of durable competition, not evidence of a fifty-fifty market. Jon Peddie Research's add-in-board reporting includes NVIDIA, AMD and Intel, with asymmetric shares. Personal AI also differs from gaming in memory requirements, processor integration and the user relationship. Its market cannot be inferred by copying graphics-card shares.[95]

NVIDIA can sell greater execution certainty

NVIDIA begins with the ability to make models execute reliably across layers. Its work spans accelerators, low-precision formats, runtime software and model development, while also supporting innovation from Qwen, GLM and MiniMax. TensorRT-LLM's published support and NVIDIA's co-design research describe a position broader than supplying one graphics card.[72, 96]

Spark paired with Flash Next makes this coordination concrete. A sparse architecture suited to large memory lets an existing device carry work that previously fit it poorly. What users may pay for is confidence that the next good model will also work on their system without patch hunting, environment reconstruction and fear of breaking a workflow.

Nemotron adds internal feedback. An in-house model team can consider state, operator shapes, low-concurrency efficiency and tool tasks before architecture is fixed. It need not win every task to have strategic value. But the essay's recorded AA configurations still put Nemotron 3.5 Lightning at 14 against Qwen3.8-27B xhigh at 34. Owning models does not replace quality and is not an automatic moat.[20, 97, 98]

I expect NVIDIA's stronger opportunity to be an experience premium in demanding personal computation. Larger models, complex workflows, reliable concurrency and development tools can support that premium. It can exist in stations, workstations, RTX computers and partner devices without requiring a Spark in every home. Dependable updates that keep older devices useful can influence the next normal computer purchase as well as today's specialist sale.

AMD can make sufficient intelligence ordinary

AMD's opportunity begins at the other end. Ryzen AI Max joins a CPU, integrated GPU and substantial unified memory in a general-purpose device. Manufacturers such as Framework turn that shape into computers for work and entertainment. Compact Qwen models through open execution software let intelligence become one capability in a computer purchase rather than a reason to add a separate accelerator.[57, 74, 99]

Once a compact MoE completes the user's work, the value of more model capacity or a more expensive execution environment has to be demonstrated. AMD need not beat NVIDIA at every performance peak to serve that demand. It needs more normally purchased configurations to be sufficient and a dependable route for developers to deliver applications on them.

This path still requires engineering. Openness does not automatically make a product easy to use. Distribution across manufacturers can produce driver, memory-allocation and runtime variation. Users will not tolerate unlimited maintenance merely because an ecosystem is open. If AMD and its partners make installation, compatibility and recovery reliable, a smaller incremental configuration cost can be more persuasive than the highest benchmark result.

Apple does not leave the market because the analogy mentions iOS

Apple is the most direct existing competitor on the default-capability route. It already connects devices, system permissions, development interfaces and customers. Foundation Models and continuing on-device model development let applications supply private assistance on hardware users already buy. More demanding functionality can draw on its cloud architecture while the relationship begins at the system interface.[78, 94, 100]

The entire personal AI market therefore cannot be partitioned using NVIDIA and AMD graphics or station shipments. The two-pole prediction primarily concerns open PCs and independently deployed personal nodes. Apple represents a separate vertically integrated channel with consumer distribution already in place. Many of its users may never feel that

they have chosen a separate AI-computing platform. They may simply continue using their phones and computers.

I do not expect a single obvious winner across all these layers. NVIDIA can earn demanding, higher-value execution work. AMD can spread adequate computation through more general-purpose configurations. Apple can preserve default user relationships within its ecosystem. Shares and profits can differ greatly while competition persists. Model suppliers and application developers cross these paths and retain part of the value.

Platforms must earn relationships, not take custody of private life

Progress by Qwen, GLM and MiniMax can improve every compatible device and reduce the cost of moving between hardware platforms. Models are both complements to hardware and a force compressing its premium. NVIDIA must keep earning its execution advantage; AMD must turn open distribution into a reliable experience. Applications such as Perplexity may hold the most immediate interface and payment relationship.[93, 96]

Durable retention comes from not rebuilding one's working life repeatedly. Files, permissions, tools and habits continue, and updates keep hardware useful. Users should also be able to export memory, replace a model and migrate original material. Model-specific KV cache may need reconstruction without making the underlying records or workflow captive. A useful platform makes staying easier, not leaving impossible.

This is an economic constraint as well as a design principle. A premium between qualifying devices must be explained by capability, reliability or time saved. More portable models weaken the ability to charge merely for access to a closed interface. Conversely, a runtime that consistently avoids trouble can sustain voluntary loyalty even when the data remains portable.

The investment account must follow normal replacement

Personal AI adoption does not turn all related computer revenue into new AI revenue. A computer someone would buy anyway already supports chip and system businesses. Incremental AI value comes from winning share, supplying more useful silicon content, providing additional services or retaining the next purchase. Renaming the entire PC replacement market as new demand and crediting it to one supplier overstates the opportunity.

A stable base of thirty million computers with a six-year replacement interval implies five million replacements annually. If AI adds USD 30, 100 or 200 of attributable silicon revenue to each replacement, incremental hardware revenue is USD 150 million, 500 million or one billion. This differs from the USD 10 billion of whole-system revenue at a USD 2000 system price and differs again from profit. The numbers are scale scenarios, not shipment forecasts.

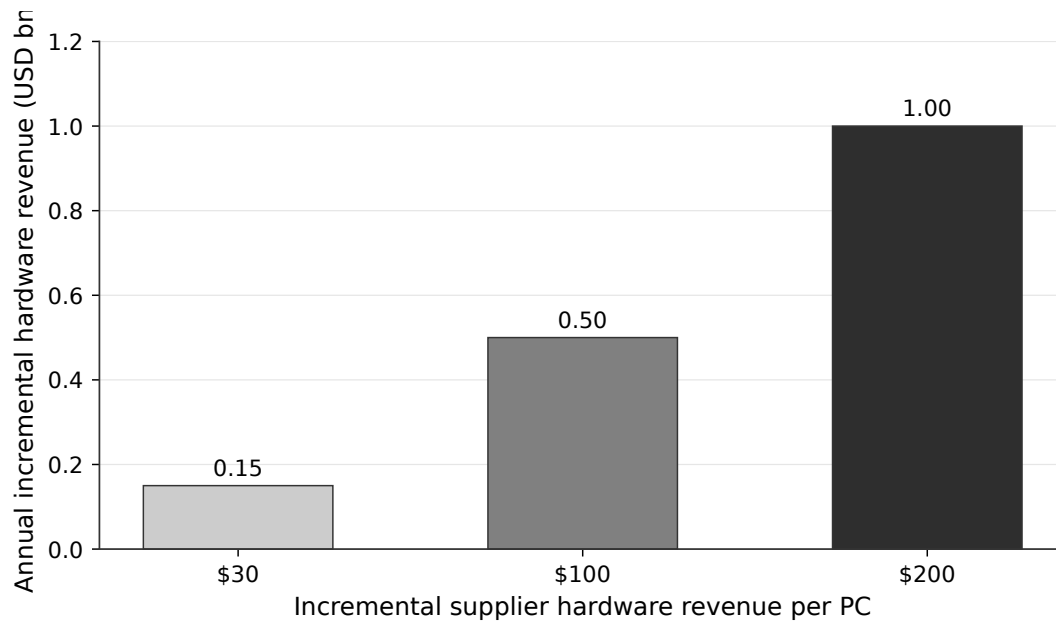


Figure 10. Computer sales are not the same as incremental AI supplier revenue.

A stable thirty-million-device base with six-year replacement. Incremental hardware content is assumed at USD 30, 100 or 200 per unit. Existing computer renewal is not counted again as an entirely new AI market. No NVIDIA or AMD share is forecast.

Useful life must be consistent on both sides. Six rather than four years makes ownership easier to amortize but reduces annual replacements in that stable base from 7.5 million to five million. Durability can attract more buyers and improve retention, but those compensating effects must occur. Growing markets require purchase cohorts, not an assumption that a recently formed installed base already produces mature replacement demand.

My investment conclusion is therefore a two-path thesis rather than a single-winner thesis. NVIDIA's advantage is more likely to appear in demanding tasks, reliable execution and higher-value ecosystems. AMD's incremental opportunity is more likely to come from penetration and useful content in general-purpose computers. Apple's control of the user entry point limits what either can capture exclusively. Personal AI can support several valuable businesses without giving them equal profits or making any current share price attractive.

The evidence worth following is the share of ordinary configurations passing stable tasks, actual activation, retention after years of use, time to dependable new-model support, and cash remaining after support and reinvestment. My base picture is sustained specialization and competition. It would be challenged by open execution remaining persistently unreliable, one closed ecosystem establishing overwhelming user lock-in, or local features failing to reduce the real burden of work.

Personal AI need not belong to a single new Apple. Integrated experience and broad distribution can grow together, giving people several durable ways to own useful intelligence.

Return to the computer that would be bought anyway

Return to the question still waiting to be sent. You want help without first disclosing the whole background. In the future I envision, those needs need not remain in conflict. A nearby system can first understand original material, preserve state and complete the part it can handle. When stronger outside intelligence is useful, you decide what to disclose.

Faster chips are only one cause of that change. Small models become more useful within similar deployment ranges. New architectures reduce the history that must be stored and read. Runtime improvements reach hardware already sold. Ordinary computer replacement brings more resources into households. Together, these changes move sufficient intelligence into computation people were already purchasing.

The first device to change everyday habits may therefore not be a dedicated AI server. It may be the next work computer, or the one already on the desk after an update adds a capability genuinely worth using. Stations can continue serving heavier work while the cloud supplies the frontier and unusual peaks. Diffusion follows the division of labor rather than requiring everybody to choose one machine.

The accounts then become clearer. An additional device must justify its complete value. A computer that would be bought anyway needs to justify only what local intelligence adds. Assets can remain useful, software can extend that usefulness, and inexpensive cloud services continue competing. Individuals need not beat datacenter production efficiency to have a reason to keep some work nearby.

Nor does this create only one winner. NVIDIA's coordinated execution, AMD's broad distribution and Apple's system interface can each sustain long-lived relationships. They compete on capability, price, reliability and control. The retention worth building lets users leave while giving them reasons to stay.

The future I foresee is not one in which everyone owns the strongest model. Personal intelligence becomes an ordinary capability of the computer. You decide what it sees, what it remembers, when it stops and when it asks for help elsewhere.

Sending remains natural. But it is no longer the first permission that must be surrendered before intelligence becomes available. It is a choice still in your hands after the work has already begun.

A. Dated evidence and comparable measurements

The report is a research perspective supported by public observations and reproducible scenarios, not a new hardware benchmark. Model and hardware records retain their original September 2026 collection dates. Additional product and platform sources were checked for this revision. A quoted configuration, a measured execution path and a task-quality result must not be combined into a fictitious single experiment.

Model lineage

The observation window contains general-purpose models near 25B to 40B total parameters. AA Intelligence Index v4.3 supplies a common retrospective ruler. Entries marked as estimates remain distinct from evaluated entries; a blank is not zero. Reasoning effort, total parameters, active parameters and architecture can all differ. The record does not independently validate a local quantized checkpoint or a 400-token completion budget.[15]

Model	Release	Total / ac- tive B	Mode	v4.3	Evidence
Qwen1.5-32B Chat	24-04-02	32	NR	NA	NA [11]
Qwen2 family	24-06-07	NA	NA	NA	NA [101]
Qwen2.5-32B Instruct	24-09-19	32.5	NR	7E	E [102]
QwQ-32B Preview	24-11-28	32	R	NA	NA [103]
QwQ-32B	25-03-05	32.8	R	9E	E [104]
Qwen3-32B	25-04-28	32.8	R	7	A [16]
Qwen3-32B	25-04-28	32.8	NR	7E	E [105]
Qwen3-30B-A3B	25-04-28	30.5/3.3	R	8E	E [106]
Qwen3-30B-A3B	25-04-28	30.5/3.3	NR	7E	E [107]
Qwen3-30B-A3B-2507	25-07-29	30.5/3.3	NR	8E	E [108]
Qwen3-30B-A3B-2507	25-07-30	30.5/3.3	R	10	A [17]
Qwen3.5-35B-A3B	26-02-24	36/3	R	19E	E [109]
Qwen3.5-35B-A3B	26-02-24	36/3	NR	15E	E [110]
Qwen3.5-27B	26-02-24	27	R	23E	E [111]
Qwen3.5-27B	26-02-24	27	NR	19E	E [112]
Qwen3.6-35B-A3B	26-04-16	36/3	R	19	A [18]
Qwen3.6-27B	26-04-22	27	R	22	A [19]
Qwen3.6-27B	26-04-22	27	NR	20E	E [113]
Qwen3.8-27B	26-08-14	27	NR	22E	E [114]
Qwen3.8-27B	26-08-14	27	xhigh	34	A [20]
Qwen3.8-27B	26-08-14	27	low	26	S [24]
Qwen3.8-27B	26-08-14	27	medium	28	S [24]
Gemma 2 27B	24-06-27	27	NR	NA	NA [115]
Gemma 3 27B	25-03-12	27.4	NR	5	A [21]
Gemma 4 31B	26-04-02	31	R	15	A [22]

Model	Release	Total / ac- tive B	Mode	v4.3	Evidence
Gemma 4 26B A4B	26-04-02	27/4	R	17E	E [23]

A. AA page not marked estimated; E. explicitly estimated; S. release summary; NA. comparable data not recovered. A does not mean independently rerun here. R denotes reasoning, NR non-reasoning; xhigh/low/medium are effort settings. The earlier four-record descriptive linear fit remains in the accompanying archive. Its approximately 19.9 index-points-per-year slope is not used to forecast intelligence or an adoption date. AA scores are not a ratio scale. Adding a compact MoE observation broadens the candidate set but does not establish a continuous progress law. Different collection dates within the same named index version remain attached to their records.

Hardware tiers and launch configurations

Tables record launch specifications, not verified present inventory. Maximum capacity must not be paired with a base price belonging to another configuration. Shared memory is not automatically fully accelerator-allocatable. Chip power limits, board power and measured wall power describe different quantities.

Chip	Announced	GB/s	Max GB	Base GB	Source
M1	20-11-10	NA	16	8	[116]
M2	22-06-06	100	24	8	[117]
M3	23-10-30	100	24	8	[41]
M4	24-10-30	120	32	16	[118]
M5	25-10-15	153	32	16	[119]
M1 Pro	21-10-18	200	32	NA	[39]
M1 Max	21-10-18	400	64	NA	[39]
M2 Pro	23-01-17	200	32	NA	[40]
M2 Max	23-01-17	400	96	NA	[40]
M3 Pro	23-10-30	150	36	NA	[120]
M3 Max	23-10-30	400	128	NA	[120]
M4 Pro	24-10-30	273	64	NA	[42]
M4 Max	24-10-30	546	128	NA	[42]
M5 Pro	26-03-03	307	64	NA	[43]
M5 Max	26-03-03	614	128	NA	[43]
M1 Ultra	22-03-08	800	128	64	[121]
M2 Ultra	23-06-05	800	192	64	[122]
M3 Ultra	25-03-05	819	512	96	[123]
M6*	26-08-25	170	NA	NA	[124]
M5 Ultra*	26-08-25	1200	512	NA	[125]

* Announced only; planned availability 2026-09-22; excluded from shipped baseline. Capacities describe launch configurations, not current inventory. M3 Ultra's launch release listed 512GB; the support page retrieved on 2026-09-12 no longer lists that option. Base configurations refer to the selected MacBook Pro or Mac Studio, not the whole PC market.

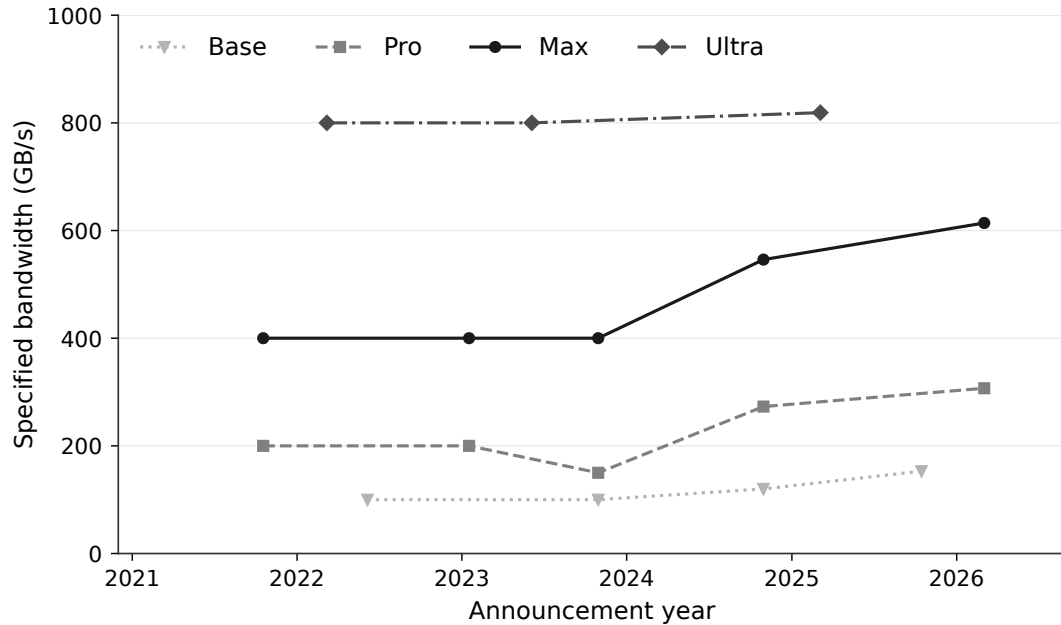


Figure 11. Bandwidth advances within product tiers rather than along one universal line.

Shipped observations in the inherited manufacturer dataset. Separate series preserve base, Pro, Max and Ultra positions; an expensive tier is not a later generation of a cheaper one. Announced-only records remain in the table but not this plot.

GPU	Launch	GB	GB/s	W	Launch USD
RTX 3080	2020-09	10	760	320	699
RTX 4080	2022-11	16	716.8	320	1199
RTX 5080	2025-01	16	960	360	999
RTX 3090	2020-09	24	936	350	1499
RTX 4090	2022-10	24	1008	450	1599
RTX 5090	2025-01	32	1792	575	1999

GPU specifications follow the manufacturer family pages and archived board records. The six entries are desktop base models, excluding Ti, SUPER and notebook variants. Original launch dates with month precision remain month precision; the analysis does not claim that a first-of-month plotting coordinate is the precise release date.[33–38]

Descriptive fits and historical checks

Within each tier, the inherited hardware fit is

$$\log y_g(t) = \alpha_g + \beta_g(t - 2020) + \epsilon_g(t), \quad \text{annualized change} = \exp(\beta_g) - 1.$$

CPU observations use one retrospective benchmark version and the stated processor configuration. They are public aggregates, not controlled for cooling, operating system or power mode. CPU performance is not converted directly into token throughput.[44, 45]

Processor	Release	Cores	Single	Multi	Source
M1	20-11-10	8	2,162	8,289	[44]
M2	22-06-06	8	2,362	9,662	[44]
M3	23-10-30	8	2,755	11,461	[44]
M4	24-10-30	10	3,237	15,204	[44]
M5	25-10-15	10	3,640	17,945	[44]
M1 Max	21-10-18	10	2,215	14,183	[44]
M2 Max	23-01-17	12	2,494	17,033	[44]
M3 Max	23-10-30	16	2,825	24,630	[44]
M4 Max	24-10-30	16	3,378	27,891	[44]
M5 Max	26-03-03	18	3,732	35,122	[44]
M1 Pro	21-10-18	10	2,196	13,863	[44]
M2 Pro	23-01-17	12	2,433	16,719	[44]
M3 Pro	23-10-30	12	2,803	16,821	[44]
M4 Pro	24-10-30	14	3,374	24,892	[44]
M5 Pro	26-03-03	18	3,692	33,599	[44]
Ryzen 7 3700X	19-07-07	8	1,589	8,957	[45, 126]
Ryzen 7 5800X	20-11-05	8	2,039	11,097	[45, 127]
Ryzen 7 7700X	22-09-27	8	2,389	13,702	[45, 128]
Ryzen 7 9700X	24-08-08	8	3,004	17,661	[45, 129]

All scores are retrospective Geekbench 7 aggregates at retrieval, not the original benchmark version at launch. OS, power mode and system price are not controlled.

Series	n	Annualized fit	95% slope interval (%)	R^2
BW / Base	4	13.4%	[-6.1, 37.0]	0.804
BW / Pro	5	12.0%	[-11.2, 41.3]	0.446
BW / Max	5	11.7%	[0.9, 23.7]	0.800
BW / Ultra	3	0.8%	[-3.8, 5.7]	0.829
CPU 1 / Base	5	11.5%	[7.6, 15.6]	0.969
CPU 1 / Pro	5	13.7%	[9.0, 18.5]	0.969
CPU 1 / Max	5	13.5%	[9.7, 17.4]	0.979
CPU 1 / AMD 8-core	4	12.7%	[6.9, 18.8]	0.979

Series	n	Annualized fit	95% slope interval (%)	R^2
CPU N / Base	5	17.3%	[10.5, 24.6]	0.960
CPU N / Pro	5	23.0%	[11.4, 36.0]	0.936
CPU N / Max	5	24.1%	[14.4, 34.7]	0.959
CPU N / AMD 8-core	4	13.9%	[11.1, 16.8]	0.996

Model. $\log(y) = \alpha + \beta(t - 2020)$; annualized change is $\exp(\beta) - 1$. Intervals describe parameter uncertainty conditional on this specification, excluding future market, software, architecture and product-policy changes.

Held-out point	n	Actual	Trend	Trend error	Hold-last error
M5	3	153	123	-19.4%	-21.6%
M4 Pro	3	273	142	-48.1%	-45.1%
M5 Pro	4	307	242	-21.3%	-11.1%
M4 Max	3	546	400	-26.7%	-26.7%
M5 Max	4	614	559	-9.0%	-11.1%
M4	3	3,237	2,947	-9.0%	-14.9%
M5	4	3,640	3,442	-5.4%	-11.1%
M4 Max	3	27,891	30,001	+7.6%	-11.7%
M5 Max	4	35,122	39,290	+11.9%	-20.6%
Ryzen 7 9700X	3	17,661	17,659	-0.0%	-22.4%

CPU Base M4/M5 rows use single-core scores; Max uses multi-core. Bandwidth is in GB/s. Error=(prediction/actual-1). Full backtests are provided in CSV. Small samples, model effort changes and product-tier decisions dominate long extrapolations. The NVIDIA three-point bandwidth regressions describe roughly 5.6% annual change for the 80 tier and 16.3% for the 90 tier. A fit using only the first two generations substantially underpredicts the third. These results are evidence against treating a fitted exponential as a product roadmap.

What the present execution records establish

Framework's compact-model comparison is a short-input, single-stream record on a specified software path. It does not pin the GMKtec 64GB retail system as the tested machine. The separate long-input community measurement uses a 128GB Strix Halo configuration. Input-only and generation-only commands are not a combined long-context generation measurement.[52, 54, 55, 79, 80] Spark measurements similarly combine distinct observations only as evidence about individual constraints. Three short-input streams at approximately 91.9 token/s aggregate, cold-input timings and longer sustained-use reports are not a single certified three-stream 140K workload. The report performs no such replay.[61, 62]

B. Capacity, work and service requirements

A fixed task set and a fixed request trace answer related but different questions. The task set tests whether the user receives the same accepted result, with limits on retries, human intervention and disclosure. The trace tests a specified computational service. A redesigned agent may use less context to finish the same task, but retrieval, tool execution, additional calls and outside advice must remain in its total cost.

A demanding reference call

The shared infrastructure case uses $I = 140000$ input tokens, resident hit fraction $h = 0.95$, and $O = 400$ actual generated tokens. These create a $hI = 133000$ -token reusable prefix and $U = (1 - h)I = 7000$ new input tokens. Counts are decimal. They are a constructed workload informed by AgentX’s public distributions, not three observed joint medians. Unseen reasoning cannot be assumed absent.[25–27] One, three and eight streams retain independent histories and receive suffixes simultaneously. The quality criterion is task-based and must not be raised merely because a newer frontier model appears. The service target is ten seconds to the first generated token, twenty seconds to complete the call, and at least thirty generated tokens per second per active stream. A loaded server has a separate two-minute allowance for a single cold 140K input. Neither this heavy target nor eight simultaneous requests describes every personal user.

Memory follows architecture and implementation

For L_f full-attention layers, n_{kv} KV heads, head dimension d and b bytes per element, the raw KV payload is

$$M_{KV}(T) = 2L_f n_{kv} d T b.$$

For the compact published Qwen3.6-35B-A3B configuration, $L_f = 10$, $n_{kv} = 2$ and $d = 256$. BF16 therefore adds 20 KiB per token. For $T = 140400$, one stream uses 2.678 GiB of raw KV. Thirty recurrent layers require state rather than a per-token full-attention KV array.[14, 51] The implementation budget uses a rounded 22.4GB file as a weight proxy, not loaded-memory measurement. One FP32 recurrent-state copy is approximately 0.06134 GiB in the inherited configuration calculation. Five copies per stream, a 5% allocation allowance, 4 GiB device workspace and 12 GiB host reserve are explicit scenarios.[52]

Streams	KV type	KV GiB	State GiB	Device GiB	Total GiB
1	1 byte	1.34	0.31	27.63	39.63
1	BF16	2.68	0.31	29.04	41.04
3	1 byte	4.02	0.92	31.09	43.09
3	BF16	8.03	0.92	35.31	47.31
8	1 byte	10.71	2.45	39.73	51.73
8	BF16	21.42	2.45	50.98	62.98

The one-byte KV cases are hypothetical storage sensitivities, not assertions that every backend supports them at equal quality. Actual allocator limits, graph buffers, draft state, host tools and thermal constraints can bind before nominal system capacity. AMD’s VGM examples distinguish total system memory from graphics allocation; other execution modes require their own checks.[53, 74]

Attention work and concurrent expert selection

A full-attention suffix must still attend to its resident prefix. With $H = hI$, $U = I - H$, query heads n_q and head dimension d , counting multiply and add as two operations gives approximately

$$F_{\text{QK+AV}} = 4L_f n_q d \left(HU + \frac{U(U+1)}{2} \right).$$

The compact reference uses $n_q = 16$ and gives about 156.5 TFLOP for this suffix. The 27B reference is about 375.7 TFLOP. These are attention operation counts only, excluding projections, experts, recurrence, movement and scheduling. They cannot be divided by a sparse FP4 advertised peak to obtain end-to-end latency.[47, 51] Selective attention changes the expensive detailed-access set while still requiring an index and retained state. It must be modeled separately. MoE activated parameters similarly measure neither all resident weights nor all bytes read by a concurrent batch. If each request selects k among E equally likely experts independently, the expected distinct count in a batch of B is $E[1 - (1 - k/E)^B]$. Real routing is correlated and unbalanced; this identity illustrates why a single-token active count is not a constant batch footprint.[59, 60]

Reverse the service requirement instead of inventing a date

Let the long-history generation rate per stream be r_d , non-input overhead be t_o , first-token limit be L_1 and completion limit be L_c . A sequential phase budget leaves

$$\tau = \min \left(L_1 - t_o - \frac{1}{r_d}, L_c - t_o - \frac{O}{r_d} \right), \quad P_{\text{suffix}} \geq \frac{BU}{\tau}, \quad \tau > 0.$$

With $t_o = 1$ second, the thresholds in the main text follow. They require the corresponding aggregate decode rate Br_d at the same history depth. Interleaved scheduling, contention, cache eviction and tail latency still need request traces. The equation is a necessary phase-budget target, not a device benchmark or a general scheduling theorem.

Streams	Decode / stream	Input seconds	Suffix token/s	Total decode
1	30	5.667	1,235	30
3	30	5.667	3,706	90
8	30	5.667	9,882	240
1	50	8.980	780	50
3	50	8.980	2,339	150
8	50	8.980	6,236	400

A candidate must pass all required gates on one available configuration. In notation,

$$C^*(t, w) = \min_{(m, d, s) \in \mathcal{A}_t(w)} C(m, d, s; t),$$

where $\mathcal{A}_t(w)$ includes only available model, device and runtime combinations meeting the task, service and disclosure conditions. If no candidate is fully verified, the minimum is unknown rather than a measured market price. Earlier restricted Spark price scenarios remain in the data archive for audit; they are not V18’s market-wide arrival forecast.

Four-bit KV quality

The cited Qwen3-30B-A3B-Instruct-2507 MRCR comparison reports BF16 KV AUC 45.8% and a protected K4V4 configuration at 42.3%, aggregated over 8K through 256K. It is not a 256K single-point loss and not a speed comparison. A separate RULER study over shorter lengths reports much smaller differences.[70, 71]

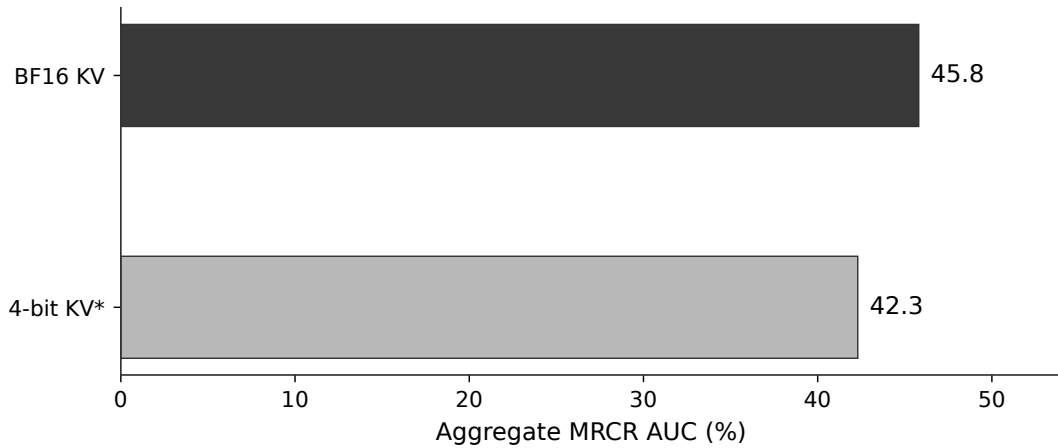


Figure 12. A smaller history representation must preserve the work it is meant to support.

Author-reported MRCR aggregate over 8K to 256K. The four-bit configuration includes protection and uses BF16 attention computation. No universal monotonic length-loss curve or local replication is inferred.

Long input and long generated reasoning must also remain separate. Quantization can change task reliability; longer reasoning additionally changes elapsed generation and future context. A smaller KV cache does not eliminate the time needed to generate many intermediate tokens. Joint weight, activation, KV and runtime optimization should be evaluated by accepted-task cost, not multiplied peak speedups.[63–65]

C. The paired-purchase economic model

The principal comparison is between two complete feasible arrangements for the same person. One obtains qualifying AI remotely while purchasing the computer required for other work.

The other adds local capability to an acceptable purchase or adds a separate device. Costs common to both arrangements cancel. The counterfactual is not the cheapest terminal regardless of its suitability and not the buyer's past spending treated as an immutable entitlement.

Capital with lifetime and a terminal value

For purchase K , net resale S after n months and annual real discount rate r_a , let

$$r = (1 + r_a)^{1/12} - 1, \quad a_n(r) = \frac{r}{1 - (1 + r)^{-n}}, \quad A(K, S, n) = \left(K - \frac{S}{(1 + r)^n} \right) a_n(r).$$

Use $a_n(0) = 1/n$ at zero interest. Capital recovery includes the opportunity cost of finance; additional depreciation or interest must not be added again. Net resale deducts selling costs. Retained service after the sale cannot also accrue to the seller.[81] If the purchase dates, horizons and residual treatment coincide, added capital recovery is

$$\Delta A = A(K_L, S_L, n) - A(K_C, S_C, n).$$

With zero residual on both sides, this equals $A(K_L - K_C, 0, n)$. It is valid for the USD 200 increment but not for arbitrary unequal holding periods. A replacement brought forward specifically for AI must account for its earlier cash outflow and the forgone use or sale of the old machine.

Years	Net residual	USD 2000	USD 4699
3	0%	62.42	146.65
4	0%	48.56	114.10
6	0%	34.79	81.75
6	20%	30.41	71.45
8	0%	27.99	65.76

For different timing or replacement plans, compute net present cost for each arrangement over a common horizon, including all purchases, sales and operating differences, then subtract. A consistent terminal value or continuation treatment prevents a plan from appearing cheaper merely because its next replacement lies just beyond the chosen horizon.

Incremental operating expenses

For the shared-PC scenario, let setup take s hours, extra maintenance be m hours per month, time value w , extra monthly application fee a , and repair allowance ρ on added hardware investment ΔK . If incremental active power is ΔP kW, electricity is e per kWh and allocated device time is t_b seconds per local call,

$$F = \Delta A + s w a_n(r) + m w + a + \frac{\rho \Delta K}{12} + F_{\text{availability}}, \quad v = \Delta P e \frac{t_b}{3600}.$$

The shared-PC base assumes the machine would already be powered during the relevant use, so $F_{\text{availability}} = 0$. This is not a claim that continuous overnight service is free. Dedicated availability instead uses idle power P_i for H hours and $v = (P_a - P_i)et_b/3600$. Each concurrent batch's device time must be counted once, then allocated across its calls.

Common assumption	Value
Holding period and residual	6 years, zero residual
Annual real capital rate	8%
Initial setup and extra maintenance	4 hours; 0.25 hours per month
Owner-time value and extra application fee	USD 50/hour; USD 0/month
Annual repair allowance	2% of added investment
Electricity and allocated device time	USD 0.20/kWh; 20 seconds/call
Shared PC added active power	100W
Dedicated system idle / active / availability	40W / 200W / 730 hours

Table 11. Explicit accounting scenarios rather than typical-owner measurements.

Crowding, battery degradation, extra failures and correction are separate costs $J(N)$. They are zero only in the baseline illustration, not necessarily in actual use. Arbitrary allocation of total past spending is avoided. For an already-owned machine, the relevant alternative can include foregone resale and other work, even though the historical invoice is sunk.

Eligible work and the avoided bill

Let N be total candidate calls, q the fraction reliably retained locally, p the eligible hosted cost per call, and ΔH the difference in adviser, retry and correction expense. The static incremental comparison is

$$\Delta C = F + qNv + J(N) + \Delta H - qNp.$$

With $J = \Delta H = 0$ and $p > v$, the break-even qualified fraction is $q^* = F/[N(p - v)]$. A fraction above one means the fixed burden cannot be recovered at that volume in this scenario. The fraction must count tasks passing the quality and disclosure conditions, not merely requests routed to a local model.

Purchase difference	Monthly USD	SG break-even	VA break-even
Same PC	17.65	4,909	5,930
USD 200 increment	21.46	6,081	7,345
USD 500 increment	27.18	7,838	9,467
USD 2000 device	62.61	18,803	22,811
USD 4699 station	114.07	34,941	42,390

The personal estimate of 500 daily calls through incompletely counted interfaces supplies $N = 15000$ under thirty-day use. It is not a population average, an audited invoice or evidence

that every call has the reference shape. At $q = 0.5$, the USD 200 increment costs USD 20.63 per month and avoids USD 25.25 at Singapore rates, or USD 21.04 in Virginia. The rest of the hosted bill remains on both sides. Continued unchanged subscriptions yield no avoided fixed fee.

Cloud caches and advisers

At input I , output O , input-token cache fraction h_c and per-million prices p_i, p_c, p_o , an implicit-cache call costs

$$p = \frac{I[(1 - h_c)p_i + h_cp_c] + Op_o}{10^6}.$$

The Singapore Flash tariff is USD 0.150 for input, 0.016 for cache reads and 0.470 for output per million tokens. Virginia uses USD 0.113, 0.014 and 0.382 respectively. Explicit write or retention fees are additional when incurred. Regional terms, model mapping and service availability must be acceptable. Local resident reuse and cloud billing hits are different measured quantities.[85, 86]

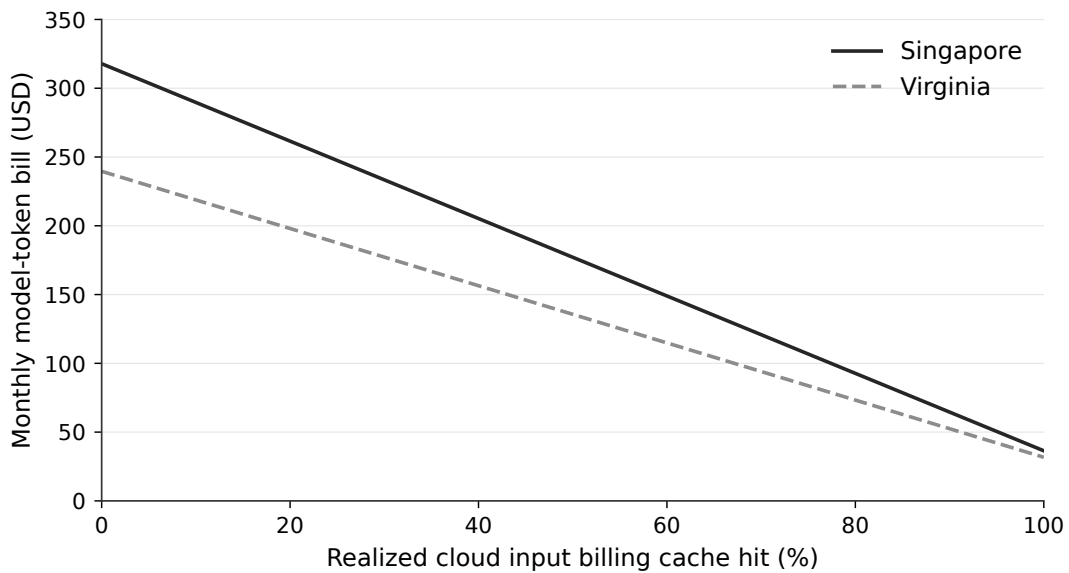


Figure 13. The cloud comparison depends on its actual billing reuse.

15000 constructed calls with 140K input and 400 output each, using the dated regional Flash rates. This is a tariff sensitivity, not a claim about either endpoint's achieved hit rate.

Identical adviser calls cancel only when both alternatives need them and prompts, frequency, reasoning and remote cache state match. Additional local failures or cold remote reconstructions do not cancel. Nor is a more expensive proprietary model automatically the appropriate avoided price. The comparator is the least expensive complete workflow meeting the task, disclosure and service requirements.

An owned device and a changing alternative

For a machine purchased now, let monthly noncapital fixed burden be R , eligible local work qN and cloud price p_m . Over the holding horizon,

$$\text{NPV}_{\text{saving}} = -(\Delta K + sw) + \frac{\Delta S}{(1+r)^n} + \sum_{m=1}^n \frac{qN(p_m - v) - R - J_m - \Delta H_m}{(1+r)^m}.$$

For a dedicated new device, replace ΔK with its full additional price and include its idle cost in R . The archived calculation varies cloud price as $p_m = p_0(1 - g_c)^{m/12}$ for $g_c = 0, 0.2, 0.4$. These are sensitivities, not forecasts. The buyer does not receive cheaper successor hardware for free, while improvements delivered to the purchased machine can change service eligibility and operating effort. New private work that would not otherwise have been purchased can be valuable, but is not avoided expenditure. A user may state a willingness to pay for control, or impose a disclosure constraint that excludes a service. No universal dollar value is assigned to privacy. The consumer choice concerns actual added burdens and benefits, while the social resource cost of producing the bundled hardware still exists.

D. Diffusion, platform value and tests of the thesis

The main industrial conclusion is a forecast of differentiated coexistence. It is neither an empirical division of the entire personal AI market into two firms nor a claim about present stock mispricing. NVIDIA's integrated execution route, AMD's broad PC distribution route and Apple's existing device-system route can all benefit from local capability while retaining different parts of the economic value.

Normal replacement and active use

Let E_t be the capable fraction of a stable computer population, λ the annual replacement fraction, s_t the capable fraction of new purchases and u_t the fraction of remaining incapable devices made capable through software. A simple recurrence is

$$E_{t+1} = (1 - \lambda)E_t + \lambda s_t + u_t(1 - \lambda)(1 - E_t).$$

The main plot sets $E_0 = 0$, $s_t = 1$, $u_t = 0$ and $\lambda = 1/L$ for $L = 4, 6, 8$. Thus $E_t = 1 - (1 - 1/L)^t$, giving about 66.5% at six elapsed years for $L = 6$. This is uniform annual replacement with a constant hazard, not a claim that every computer is retired exactly at age six. A deterministic six-year cycle with a different initial age distribution gives a different path. No curve predicts a calendar date or measured household replacement distribution. Active use can be written $U_t = a_t E_t$, with activation a_t measured separately. The accompanying dataset includes an illustrative 60% activation factor, not a market estimate. Task capability, default installation, activation and repeated successful use must remain separate. Device turnover can spread a feature without any purchase motivated exclusively by AI.

Incremental hardware content and value capture

For steady-state active stock B and replacement life L , annual replacements are $Q = B/L$. If incremental attributable AI hardware revenue per unit is Δv , then

$$\Delta R_{\text{hardware}} = Q\Delta v.$$

At $B = 30$ million and $L = 6$, $Q = 5$ million. With $\Delta v = 30, 100, 200$ USD, annual incremental hardware revenue is USD 0.15, 0.5 and 1 billion. The accompanying worksheet also tests four- and eight-year replacement. These quantities are neither total PC revenue nor NVIDIA or AMD forecasts. A supplier can additionally change its share, and some customers can buy entirely new dedicated devices. Those effects must be modeled separately from increasing chip content on a computer already in the counterfactual. If Q_L, Q_C are units captured under local-capability and baseline arrangements, one valid hardware decomposition is

$$\Delta R = Q_L v_L - Q_C v_C = (Q_L - Q_C) v_C + Q_L (v_L - v_C).$$

The first term is captured-unit change and the second content change. Adding the whole $Q_L v_L$ again as an independent AI market would count the baseline twice.

Purchase cohorts and recurring services

In a growing market, current installed base divided by average life does not identify that year's replacements. Let x_t be new adopters, p_a the retirement-age distribution, and r_a the probability a retiring owner stays with the platform. With q_t all units sold, cohort replacement can be represented as

$$q_t = x_t + \sum_{a \geq 1} q_{t-a} p_a r_a.$$

Survival and activation separately determine active stock. A second-hand sale may preserve a device in the installed base without creating another new hardware sale. Longer service can improve retention and adoption while postponing replacements. These mechanisms are not interchangeable sources of revenue. For active stock B_t , paid-service attachment α , net supplier revenue p_s per attached device, contribution margins m_h, m_s and fixed platform expense F_t , an illustrative operating contribution is

$$\Pi_t = R_{h,t} m_h + B_t \alpha p_s m_s - F_t.$$

This is before tax, capital expenditure and working capital, not free cash flow. Application subscriptions do not become silicon-supplier revenue unless the contractual share is included. Human support can shift a cost rather than eliminate it; automation can create an actual joint saving.

What would validate or overturn the platform picture

The two-path thesis predicts that coordinated execution retains a paid niche while sufficiently reliable open-model deployment expands through general-purpose computers. It would gain support from comparable task tests across NVIDIA and AMD, persistent demand in both premium and ordinary tiers, multi-vendor model availability and actual long-term retention. Apple's activation and default distribution must be measured as their own route. It would be challenged if one ecosystem controlled access so strongly that the other could not remain an attractive alternative, if open execution failed to become dependable at normal maintenance cost, or if most purported local work still required substantial uncanceled remote computation. An asymmetric share alone would not disprove coexistence. No fifty-fifty outcome is assumed.[74, 78, 95, 96] An industry opportunity is also not a valuation. At a 10% discount rate and 3% perpetual growth, USD 100 billion of value for an immediately mature incremental business requires USD 7 billion of next-year free cash flow. Delayed maturity and transition investment change that requirement. This algebra does not say what investors currently price in, and operating contribution cannot substitute for free cash flow.

Replication and disclosure

The source package includes dated source records, retained original datasets, current calculation scripts and grayscale plotting scripts. `calculate_v18.py` recomputes the incremental purchase, eligible-share, cache, capital, memory, service, diffusion and supplier-content tables without running a language model. The workbook exposes corresponding assumptions and formulas. Archived restricted-date outputs are labeled historical scenario work rather than present headline conclusions. Generative AI assisted drafting, translation, code preparation and editorial revision. It is not an author. The named author is responsible for the manuscript and its interpretation. No new GPU tests, measured device-lifetime distribution, audited personal-usage trace or household adoption survey is claimed. The principal unresolved empirical test is a matched task-and-service replay across available configurations, followed by actual incremental maintenance and cloud-billing records.

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